

Packet Capture (PCAP) Forensic Investigation and Threat Analysis

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Role: SOC Analyst / Network Forensics Investigator

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Executive Summary

This investigation focused on the forensic analysis of a packet capture (PCAP) file associated with suspicious network activity involving a company executive attending a Bring Your Own Device (BYOD) conference. Using Wireshark within a VMware virtual environment, the investigation analyzed DNS traffic, HTTP communications, packet streams, file downloads, and encrypted attachments to determine whether malicious activity occurred and to identify indicators of compromise (IOCs).

The investigation revealed evidence that the executive accessed a spoofed or non-legitimate website impersonating a trusted domain. Packet analysis identified suspicious HTTP exchanges, unauthorized executable downloads, and potential malware activity originating from the fraudulent website. Additional analysis uncovered evidence of travel ticket purchases and decrypted email attachments suggesting possible interpersonal misconduct and potential social engineering or reputational targeting.

The findings demonstrate how packet capture analysis can uncover malicious activity, phishing attempts, unauthorized downloads, and insider-related concerns. The investigation highlights the importance of DNS monitoring, web traffic inspection, endpoint protection, and cybersecurity awareness training in defending against advanced social engineering and malware-based attacks.

Objective

The purpose of this investigation was to:

1. Analyze a PCAP file using Wireshark to identify suspicious network activity.

- 2. Determine whether malicious websites or spoofed domains were accessed.
- 3. Investigate HTTP traffic and downloaded files for indicators of compromise.
- 4. Confirm suspicious user activity including ticket purchases and file transfers.
- 5. Reconstruct the timeline of events leading to the incident.
- 6. Assess the potential cybersecurity and organizational risks associated with the activity.

Investigation Environment

Tools Utilized

Tool	Purpose
Wireshark	Packet capture and protocol analysis
VMware Workstation	Secure forensic analysis environment
PCAP File Analysis	Network forensic investigation

Investigation Methodology

The investigation followed a structured forensic workflow designed to analyze suspicious traffic patterns and identify indicators of compromise.

Investigation Phases

- 1. Protocol hierarchy analysis
- 2. DNS traffic investigation
- 3. Website legitimacy verification
- 4. HTTP stream analysis
- 5. File download investigation
- 6. Email attachment analysis
- 7. Timeline reconstruction
- 8. Threat assessment and reporting

Investigation Findings

Phase 1 – Protocol Hierarchy Analysis

Protocol hierarchy analysis revealed that the majority of captured traffic involved TCP communications.

Protocol Distribution

Protocol	Percentage
TCP	92.8%
HTTP over TCP	2.8%

The dominance of TCP traffic indicated extensive web-based communication activity throughout the capture period.

Phase 2 – DNS Investigation

Investigators analyzed DNS queries associated with the domain:

- www.harveycartel.org

Key Findings

Indicator	Result
Website Accessed	Yes
Related Frames	5668, 5669, 5671, 5672, 5675
Source IP Address	172.16.254.128
Source Port	56207
Total Capture Duration	Approximately 3 minutes 11 seconds

The DNS traffic confirmed that the executive's system initiated communication with the suspicious domain during the captured timeframe.

Phase 3 – Non-Legitimate Website Analysis

Further analysis revealed discrepancies between the legitimate IP address of harveycartel.org and the IP address observed in the packet capture.

Legitimate IP Verification

Observation	Result
Legitimate IP Address	207.150.212.43
Captured IP Address	Different from legitimate host
Assessment	Potential spoofed or malicious website

The evidence strongly suggested that the executive was redirected to a fraudulent or cloned website designed to imitate the legitimate domain.

Suspicious Packet Range

Suspicious Frames

5670 – 10912

These frames likely represented malicious communication with the rogue server.

Threat Assessment

Downloading content from a non-legitimate website introduced several potential risks:

- Malware infection
- Trojan deployment
- Credential theft
- Data corruption
- Ransomware delivery

- Unauthorized remote access

Phase 4 – HTTP Stream and Malware Investigation

HTTP stream analysis identified suspicious download activity originating from the rogue website.

Download Source

Indicator	Result
Download URL	http://www.thewayoftheninja.org/n.html
Downloaded File	Nv2-PC.exe

The downloaded executable file strongly indicates a potential malware payload or trojanized application.

Security Implications

The presence of an executable download from a fraudulent website suggests the system may have been compromised through:

- Cross-site scripting (XSS)
- Malicious redirection
- Social engineering
- Malware delivery

This activity represents a high-confidence indicator of compromise.

Phase 5 – Travel Ticket Investigation

Investigators identified packet evidence associated with travel ticket purchases.

Findings

Indicator	Result
Travel Purchase Confirmed	Yes
Number of Tickets	2
Destination	Dubai
Supporting Files	EmiratesETicket1.pdf, EmiratesETicket2.pdf

Relevant packet analysis identified travel-related PDF attachments embedded within network traffic.

Relevant Packets

Packet Number	Attachment
11784	EmiratesETicket1.pdf
12038	EmiratesETicket2.pdf

The travel documentation suggested coordinated travel arrangements involving the executive and another individual.

Phase 6 – Email Attachment and Insider Risk Analysis

The forensic team decrypted attachments associated with packets 11745 and 12038.

Attachment Findings

Individual Referenced	Execution Date
John Williams	April 1, 2015
Sarah Galloway	April 1, 2015

The synchronized travel information and attachment metadata suggested possible interpersonal misconduct involving the executive and another employee's spouse.

Insider Risk Indicators

The investigation identified several concerns:

- Potential abuse of organizational resources
- Possible reputational targeting
- Suspicious attachment timing
- Indicators of unauthorized monitoring
- Potential social engineering manipulation

The simultaneous creation timing of attachments suggested possible malicious automation or unauthorized surveillance activity occurring on the executive's system.

Timeline Reconstruction

Timeframe	Activity
Frame 5668–5675	DNS queries to suspicious domain
Frame 5670–10912	Communication with fraudulent website
Mid-capture	Executable file download detected
Packet 11784	Travel ticket attachment identified
Packet 12038	Second travel attachment identified
Final analysis	Evidence of suspicious interpersonal activity

The reconstructed timeline demonstrated a progression from domain access to potential compromise and eventual discovery of sensitive information.

SOC Analyst Assessment

This investigation identified strong evidence of malicious web activity involving access to a spoofed or fraudulent website that delivered a suspicious executable file to the executive's system. DNS analysis, HTTP stream inspection, and packet reconstruction collectively indicate that the user was likely exposed to malware or phishing infrastructure during the BYOD conference environment.

The downloaded executable file, Nv2-PC.exe, represents a significant indicator of compromise and suggests that the endpoint may have been infected through malicious web activity or social engineering tactics. The evidence also demonstrated how attackers may leverage spoofed domains and deceptive web content to compromise systems and collect sensitive information without immediate user awareness.

Additionally, the investigation uncovered sensitive interpersonal information through decrypted attachments and ticket purchases, highlighting the broader risks associated with insider threats, reputational targeting, and unauthorized surveillance. The combination of malware exposure and sensitive personal information significantly increased organizational risk and demonstrated the importance of endpoint monitoring, DNS inspection, and employee cybersecurity awareness.

Recommendations

Immediate Security Actions

1. Isolate and scan the affected endpoint for malware.
2. Block malicious domains and associated IP addresses.
3. Reset potentially compromised credentials.
4. Conduct endpoint forensic imaging and malware analysis.
5. Review outbound communications for additional indicators of compromise.

Security Improvements

Endpoint Protection

- Deploy advanced endpoint detection and response (EDR)
- Enable application execution controls
- Restrict unauthorized executable downloads

Network Security

- Monitor DNS anomalies
- Implement web filtering
- Inspect HTTP and HTTPS traffic
- Deploy intrusion detection systems (IDS)

User Awareness

- Conduct phishing awareness training
- Educate users on spoofed domains
- Reinforce secure BYOD practices

Incident Response

- Develop PCAP analysis procedures
- Improve malware response workflows
- Establish insider threat monitoring policies

Key Skills Demonstrated

Cybersecurity Skills

- Network Forensics
- Packet Capture Analysis
- DNS Investigation
- HTTP Stream Analysis
- Malware Identification
- Threat Hunting
- Timeline Reconstruction
- Social Engineering Analysis
- Insider Threat Assessment
- Incident Response

Technical Tools

- Wireshark
- VMware Workstation
- PCAP Analysis
- Protocol Inspection
- HTTP Object Extraction

- DNS Traffic Analysis

Conclusion

This investigation successfully reconstructed suspicious network activity involving a potentially compromised endpoint, malicious website access, executable malware downloads, and sensitive communications identified through packet capture analysis.

The forensic analysis demonstrated how SOC analysts can leverage packet inspection and protocol analysis to identify phishing infrastructure, malware delivery mechanisms, and insider-related risks. The findings emphasize the importance of proactive monitoring, secure browsing practices, endpoint protection, and cybersecurity awareness in protecting organizational systems and sensitive information.